

Andrew E. Toader

Pittsburgh, PA, USA

EDUCATION

College of Medicine, Lake Erie College of Osteopathic Medicine

July 2023 – June 2027

Doctor of Osteopathic Medicine (DO)

Swanson School of Engineering, University of Pittsburgh

August 2018 – May 2022

Bachelor of Science in Engineering (Honors) – Computer Engineering – Magna Cum Laude

Thesis: “Differences in excitatory and inhibitory neuron oxygen metabolism elucidated by intrinsic optical imaging and optogenetics in awake mice.”

Minor in Chemistry

HONORS & AWARDS

The 2024 Campagna Scholar in Neurological Surgery (\$5000) – Oregon Health and Sciences University

Summer 2024

1st Place in Medical Student Research – Lake Erie College of Osteopathic Medicine

October 2023

The 2022 Outstanding Graduate in Computer Engineering – University of Pittsburgh

May 2022

Highest Undergraduate Distinction, Bachelor of Philosophy – University of Pittsburgh Honors College

May 2022

Magna Cum Laude – University of Pittsburgh

May 2022

Summer Undergraduate Research Funding (\$4000) – Swanson School of Engineering

April 2021

Eta Kappa Nu Honors Society – Institute of Electrical and Electronics Engineers

April 2021

Best Undergraduate Research – University of Pittsburgh Rehabilitation Institute

July 2019

APPOINTMENTS & PROFESSIONAL EXPERIENCES

Research Scholar, Neurological Surgery, Oregon Health & Sciences University

May 2024 – present

Principal Investigator: Daniel Cleary, MD, PhD

Focus: Design and implement clinical tremor-detection devices and algorithms to improve diagnosis and surgical treatment of Essential Tremor and Parkinson's disease.

Rationale: Current clinical tremor evaluation relies on subjective assessment; these tools provide objective, quantifiable metrics to guide surgical decision-making and assess post-operative outcomes.

Projects/Devices:

- **Accelerometer attachment to pen:** MRI-safe tablet/pen system with motion sensors to quantify tremor during spiral drawing. (Link: https://github.com/clearylaboratory/tremor_analysis). *I was designer/creator.*
- **Image-based spiral analysis:** Algorithm to measure tremor severity from scanned paper-drawn spirals without hardware. *I was designer/creator.*
- **Augmented Reality (AR) Headset:** AR headset projecting 3D tasks (spiral drawing, object manipulation) for real-time tremor quantification. *I lead and advised a team of 3 people in creating device.*

Engineering Skills:

- Device design and prototyping
- Signal processing and frequency-domain analysis
- Algorithm development for biomedical signals
- Computer programming (Python, MATLAB)
- Hardware/software integration and bench testing
- Team leadership in engineering projects

Research/Medical Skills:

- Study design and protocol development
- IRB/ethics submission and regulatory compliance
- Data acquisition, management, statistical analysis

- Scientific writing: grants, abstracts, peer-reviewed manuscripts
- Presentation of findings at national meetings

Researcher, University of Pittsburgh NeuroImaging Laboratory

Nov 2018 – present

Principal Investigator: *Alberto L. Vazquez, PhD*

Focus: Investigate neuronal health and metabolism in disease states (ischemic disease, Alzheimer's) using optical imaging techniques.

Rationale: High-resolution optical imaging of the rodent brain allows for manipulation and exploration of neuronal health and identification of therapeutic targets in ischemic and metabolic diseases of the brain.

Projects:

- **Metabolic demand vs neuronal firing in cardiac arrest model:** Imaged neuronal metabolic function before + after experimental cardiac arrest to examine coupling of neuronal activity and metabolism. *I performed all analysis and algorithm development.*
- **Optical stimulation of implanted electrodes:** Used two-photon microscopy to stimulate optically reactive electrodes and measure neuronal firing with Ca^{2+} -sensitive fluorescence. Framework for possible future clinical translation. *I performed experiments stimulating electrodes and analyzed two-photon imaging data.*
- **Optical measure of cerebral metabolic rate of oxygen consumption (CMRO_2) in rodent cortex:** adapted a well-established MRI-based imaging computation of neuronal oxygen usage to optical imaging. *I performed all mathematical derivation, testing, and model validation.*
- **CMRO_2 in specific neuronal subpopulations:** Used our optical imaging model to compute oxygen consumption of selectively activated excitatory and inhibitory neuron subpopulations using genetic channel insertion. *I conceptualized and designed the study, carried out experiments, analyzed data, and was the project leader. This was my undergraduate thesis work.*

Engineering Skills:

- Image processing
- Biosignals (electrophysiology, optical imaging) analysis
- Optics engineering
- Computer programming (MATLAB, C)

Research/Medical Skills:

- Basic science and translational research design
- Laboratory mentorship
- Cross-disciplinary collaboration
- Inferring brain physiology from electrical and metabolic data
- Manuscript preparation
- Presentation of findings at national conferences

Student Researcher, Safar Center, UPMC Children's Hospital of Pittsburgh

August 2018 – April 2022

Principal Investigator: *Mioara Manole, MD*

Focus: Investigated effects of cardiac arrest on microvasculature of the brain.

Rationale: Persistent neurological injury after cardiac arrest is present in over 50% of pediatric patients despite adequate resuscitation. Underlying mechanisms remain poorly understood.

Projects:

- **Investigating microvascular dysfunction after epinephrine use in cardiac arrest:** Used multi-photon microscopy to image cerebral microvasculature in rodents after experimental cardiac arrest. Measured vessel diameter and blood flow in rodents resuscitated with and without

epinephrine. I performed all algorithm development and data analysis, and assisted in image acquisition during experiments.

Engineering Skills:

- Computer vision algorithm development
- Computer programming (Python, MATLAB)

Research/Medical Skills:

- Designing and executing complex experiments (8-hour experiments)
- Working in multidisciplinary research teams
- Understanding resuscitation pharmacology and associated physiological effects
- Statistical analysis
- Manuscript writing

Embedded Systems Engineering Intern, Cardiac Ablation Solutions, Medtronic

June 2020 – August 2020

Manager: Kevin Ponziani, M.Sc.

Focus: Developed commercial, FDA-approved devices to treat and diagnose cardiac conduction abnormalities.

Project:

- **Cryoablation system development:** Worked in a team of ~100 engineers to design a device that ablates heart tissue using very low temperatures to disrupt abnormal circuits in the heart. *I worked on the communication between the two computers controlling the device.*

Engineering Skills:

- Embedded systems design
- Microcontroller programming (C/C++)
- TCP/IP and UDP wireless communication protocols
- Integrated development environment (IDE) use with bench testing (IAR Embedded Workbench)
- Integration of multi-engineer codebases

RESEARCH PUBLICATIONS

1. Oghifobibi OA, **Toader AE**, Nicholas MA, Nelson BP, Alindogan NG, Wolf MS, Kline AE, Nouraie SM, Bondi CO, Iordanova B, Clark RS, Bayir H, Loughran PA, Watkins SC, St Croix CM, Kochanek PM, Vazquez AL, Manole MD. Resuscitation with epinephrine worsens cerebral capillary no-reflow after experimental pediatric cardiac arrest: An in vivo multiphoton microscopy evaluation. *J Cereb Blood Flow Metab.* 2022 Jul 19:271678X221113022. doi: 10.1177/0271678X221113022. PMID: 35854408.
2. **Toader AE**, Fukuda M, Vazquez AL. Evaluation of calibrated and uncalibrated optical imaging approaches for relative cerebral oxygen metabolism measurements in awake mice. *Physiol Meas.* 2024 Apr 24;45(4):045007. doi: 10.1088/1361-6579/ad3a2d. PMID: 38569522; PMCID: PMC11041811.
3. **Toader AE**, Fukuda MH, Vazquez AL. “Differences in excitatory and inhibitory neuron oxygen metabolism elucidated by intrinsic optical imaging and optogenetics in awake mice”. *Ingenium: Undergraduate Research at the Swanson School of Engineering.* May 2022.
4. (Accepted with revisions):
Toader AE, Useinovic N, Shafie B, Henn MC, Joyce M, Smith HD, Neilson LE, Safarpour D, Raslan AM, Cleary DC. Combined accelerometry and writing analysis for quantifying tremor during focused ultrasound thalamotomy. *J Neurosurg.*
5. (Submitted, awaiting decision):
Toader AE, Cover CG, Ruff CF, Fukuda M, Ross SE, Vazquez AL. Excitatory and Inhibitory neuron contributions to hemodynamic responses evoked by whisker stimulation in awake mice. *Imaging Neurosci (Camb).*
6. (Submitted, awaiting decision):
Cahill J, Useinovic N, **Toader AE**, Minns H, Lipson A, Cleary DC. Direct targeting for focused ultrasound thalamotomy in the treatment of movement disorders. *J Neurosurg.*

RESEARCH PRESENTATIONS

1. **American Society of Anesthesiology, San Antonio, TX – Poster (October 2025)**
“Abdominal wall nerve blocks with liposomal bupivacaine: two case reports”. **Toader AE**, Augi T, Soltys K, Visoiu M
2. **Regional Anesthesiology & Acute Pain Medicine, Orlando, FL – Poster (May 2025)**
“Liposomal bupivacaine for supraclavicular nerve block in a pediatric patient on CRRT: A case report”. **Toader AE**, Visoiu M
3. **American Association of Neurological Surgeons, Boston, MA – Poster (April 2025)**
“A novel device for quantifying tremor during focused ultrasound thalamotomy”. **Toader AE**, Useinovic N, Shafie B, Henn MC, Joyce M, Smith HD, Neilson L, Safarpour D, Raslan AM, Cleary DR
4. **BRAIN & BRAIN PET, Glasgow, Scotland – Poster/Flash Presentation (May 2022)**
“Differences in excitatory and inhibitory neuron oxygen metabolism elucidated by optical imaging in awake mice”. **Toader AE**, Vazquez AL
5. **BRAIN & BRAIN PET, Glasgow, Scotland – Poster Presentation (May 2022)**
“A Novel Method for the Assessment of Capillary Stasis (No-Reflow) After Cardiac Arrest”. **Toader AE**, Gunasekaran V, Nelson B, Wang P, Vazquez AL, Manole MD
6. **Bachelor of Philosophy Thesis Defense Presentation, Pittsburgh, Pennsylvania (April 2022)**
“Differences in excitatory and inhibitory neuron oxygen metabolism elucidated by optical imaging in awake mice.” **Toader AE**
7. **Pediatric Academic Societies International Conference, Philadelphia, PA – Poster (2020)**
“In vivo Assessment of Neuronal and Metabolic Activity in the Cortex After Pediatric Cardiac Arrest”. **Toader AE**, Iordanova B, Alexander H, Fukuda H, Li L, Clark RS, Vazquez AL, Kochanek PM, Manole MD
8. **Pediatric Academic Societies International Conference, Philadelphia, PA – Platform Presentation (2020)**
“Epinephrine worsens cortical microcirculatory blood flow after experimental pediatric cardiac arrest”. Ogofobibi O, **Toader AE**, Alindogan NG, Alexander H, Nelson BP, Bayir H, Clark R, Dezfulian C, Kochanek PM, Manole MD
9. **University of Pittsburgh Rehabilitation Institute Research Day – Podium Presentation (June 2019)**
“In vivo assessment of neural and metabolic activity after pediatric cardiac arrest”. **Toader AE**, Iordanova B, Fukuda H, Alexander H, Manole MD

VOLUNTEER ACTIVITIES

Senior Patroller, National Ski Patrol

2015 – present

Hours: 6-12 hours/week during Fall and Winter months.

Description:

- Deliver pre-hospital medical care to injured skiers and snowboarders on the slopes of ski areas in teams of other patrollers. Continually assess safety of slopes to prevent injuries
- Teach incoming candidates skiing and medical skills

Positions held:

- *Senior Patroller* (2024 – present)
- *Ski + Toboggan Instructor* (2024 – present)
- *Alpine Patroller* (2015 – 2024)

Patrols served:

- Seven Springs, PA (2017 – present)
- Boyce Park, PA (2015 – 2017)
- Mount Pleasant, PA (2024 – 2025)

Leader of Student Volunteers, Children's Hospital First Clinical Experiences

Sept. 2021 – July 2023

Hours: 8 hours/week working and directing program

Rationale: Started program to assist physicians and nurses with tasks in emergency department due to unprecedented patient volume during COVID-19 Pandemic. Program concurrently exposed undergraduate students interested in a career in medicine to a clinical setting.

Description:

- Created and directed a program of undergraduate students who assist with various tasks in the Emergency Department of Children's Hospital.
- Scheduled volunteers and communicated with the physician and nurse leadership in the Emergency Department to determine the areas and tasks where volunteers are needed.

CPR Teacher, Pittsburgh Public Schools / UPMC Children's Hospital

May 2022 – July 2023

Hours: 20 hours including instruction and planning time

Description:

- Taught hands-only CPR to middle-school aged students to increase bystander CPR in the community of Pittsburgh.

Medical Mission in Aguacate, Guatemala in Fr. Andres Giron Clinic

January 2018

Hours: 70 hours over one week

Description:

- Leader of optometry clinic. Assessed vision with portable auto-refractor. Screened 220 patients and fitted 80 patients with glasses. Assisted with medical procedures in the clinic.

Blind Skier Guide, Blind Outdoor Leisure Development (BOLD)

Jan. 2015 – present

Hours: 4-8 hours per season, available as necessary

Rationale: Help vision-impaired persons enjoy outdoor sports

Description:

- Work in a team of 3 people to guide blind skiers down the slopes using verbal commands.

ORGANIZATIONS

American Academy of Neurology

March 2024 – present

American Association of Neurological Surgeons

March 2024 – present

Congress of Neurological Surgeons

March 2024 – present

American Academy of Pediatrics

January 2024 – present

Sigma Sigma Phi Osteopathic Medicine Honors Society

January 2024 – present

- *Treasurer, LECOM Chapter (2024 – 2025)*

American College of Osteopathic Surgeons

July 2023 – present

American Osteopathic Association

July 2023 – present

Institute of Electrical and Electronics Engineers (IEEE)

April 2021 – present

- *Corporate Liaison, University of Pittsburgh Chapter (2020 – 2021)*

CERTIFICATIONS

Advanced Cardiovascular Life Support (ACLS)

May 2025

Basic Life Support (BLS) For Healthcare Providers

since 2015

Outdoor Emergency Care (OEC)

2015